Validation Evidence

AnalystLab Africa Experience Lab | Week 3 | Track A: Data Analytics | Author: Divine Okezie Nwaoha | Tools: Power BI, Power Query, DAX, Python/SQL (validation)

Independent checks were run on the raw CSVs with SQL (in-memory) and Python, then compared with the dashboard values shown in the dashboard screenshots (04_Dashboard_Screenshots.pdf). Dashboard cards and labels display rounded values (for example 28K), so "Yes (rounded)" means the figures agree at the precision the dashboard displays.

#	Metric	Independent check (SQL/Python)	Dashboard value (page)	Match?
1	Total encounters	27,891	27,891 (Page 7 decomposition tree; page header). Cards show 28K	Yes
2	Unique patients	974	974 (Pages 1 and 6)	Yes
3	Zero-coverage encounters	13,586 (48.71%)	48.71% (Page 4); 48.7% (Page 1). The count is not displayed directly	Yes (percent)
4	Encounters over 24h	75 (0.27%)	0.3% (Page 1 card); 0.27% (Page 3 text). The count is not displayed directly	Yes (rounded)
5	Ambulatory encounters	12,537 (44.95%)	44.95% class share (Page 3); 12.5K (Page 1)	Yes
6	Avg claim cost: Medicaid / Medicare	$6,205 / $2,168	$6.2K / $2.2K (Page 4)	Yes (rounded)
7	Zero Coverage Cost Share	62.2%	62.2% (Page 1)	Yes
8	Encounters 2014 / 2021	3,885 / 3,530	3.9K / 3.5K (Page 2)	Yes (rounded)

Filter test

Apply the Encounter Class slicer to "ambulatory". Total Encounters should drop to 12,537 and all cards should update. Repeat with one payer and one year, then screenshot each. **Status:** the supplied screenshots show every slicer at its default (All, 2011 to 2022), so filter-test screenshots still need to be added.

Method

The encounter, procedure and payer files were loaded into an in-memory SQL database; the reference questions (yearly volume, class share, 24h split, zero coverage, top procedures, average claim by payer, quarterly patients, 30-day returns) were run on the raw data and compared with DAX results. The supplied database script was not executed, because it drops and recreates `hospital_db`.

Discrepancy log

Differences larger than rounding between the independent checks and the dashboard.

#	Difference	Cause	Status
1	Insured Zero Coverage card (Page 4) shows $13.586K. Expected about 4,779 encounters (zero coverage excluding NO_INSURANCE).	The card shows the full 13,586 zero-coverage encounters in currency format, so the NO_INSURANCE exclusion is not applied and the number format is wrong.	Open. Fix the measure's format and filter, then re-screenshot Page 4.
2	Exact counts for zero-coverage encounters (13,586) and encounters over 24h (75) are not displayed on any page.	Dashboard shows these only as percentages (48.71% and 0.3%).	Checked by percentage only. Optionally add count cards.